

Students names:.....

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Group:

Date:

Experiment N°01(Experimental)

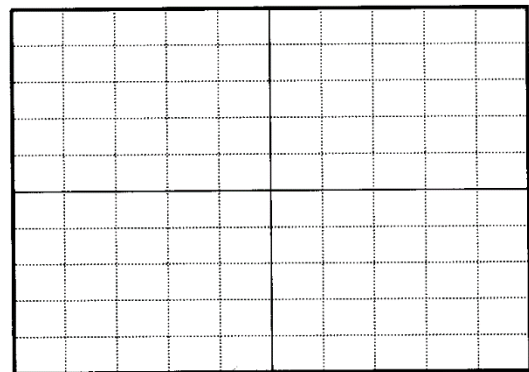
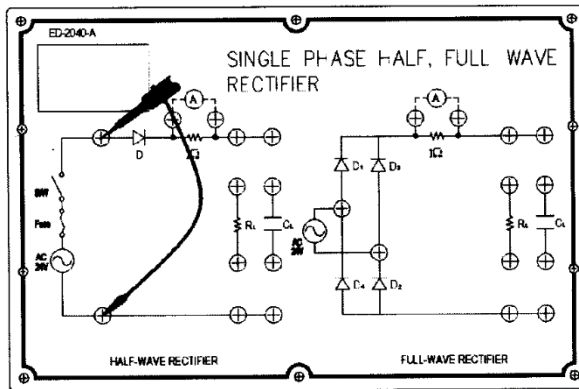
A Single-Phase Half Wave/Full Wave Rectification Circuit

Procedure :

Part one: Half wave rectifier

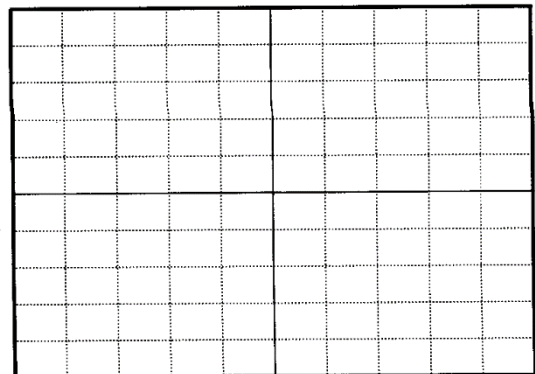
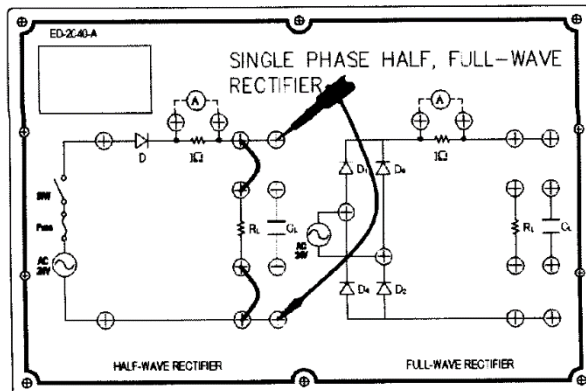
with R Load

1. Switch on the power supply.
2. Measure and display the input voltage waveform as the figure shown below.



Time_____ms/div, Volt_____V/div

3. Connect the circuit as shown in figure below.
4. Measure the load DC and AC voltage values using the oscilloscope, record the result in table I.



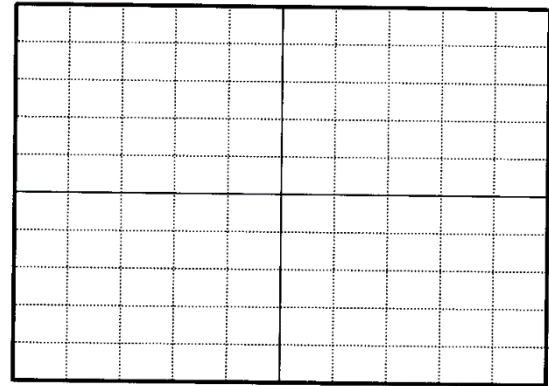
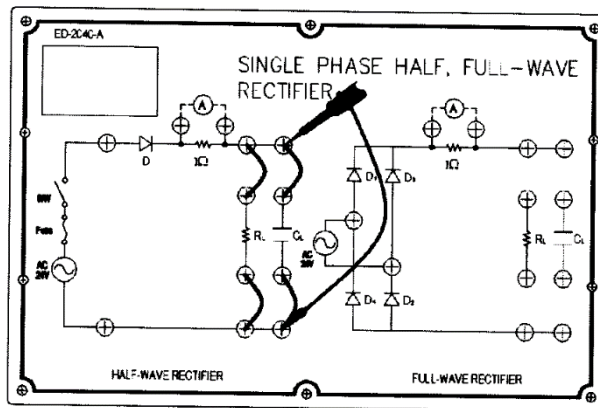
Time_____ms/div, Volt_____V/div

5. Display the load voltage and current waveforms on the oscilloscope.
6. Display the voltage waveform across the diode on the oscilloscope
7. Practically observing the waveform of current on Oscilloscope in the Lab: the connected 1Ω resistor within the path of which the current is to be observed. According to Ohm's law, the voltage waveform across the 1Ω resistor is also the waveform of the current flowing through it.
8. Plot the displayed waveforms in the graph paper.

- Switch off the power supply.

With R-C Load

- Connect the circuit as shown in the following figure.



Time_____ms/div, Volt_____V/div

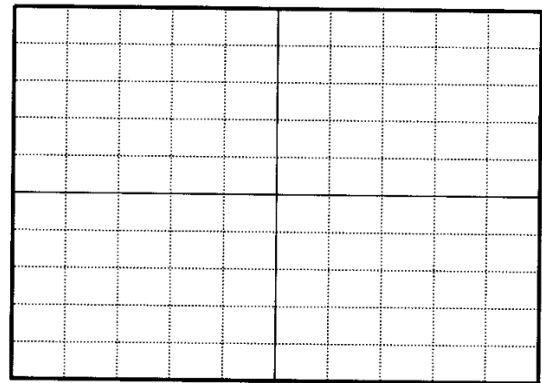
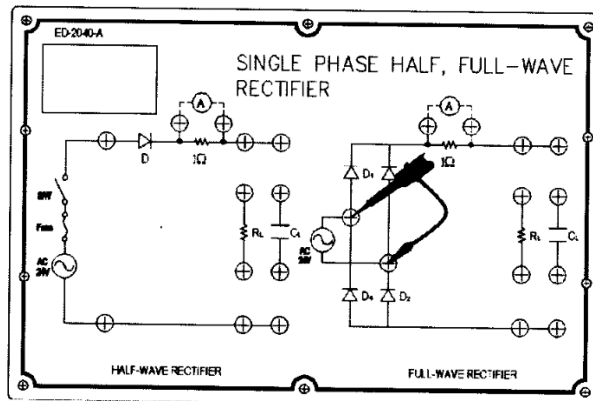
- Switch on the power supply.
- Measure the load dc and ac voltage values, record the result in table I.
- Display the load voltage and current waveforms on the oscilloscope.
- Display the voltage waveform across the diode on the oscilloscope
- Plot the displayed waveforms in your graph paper.
- Switch off the power supply.

Part two: Full wave rectifier

Repeat part one with the full wave rectifier.

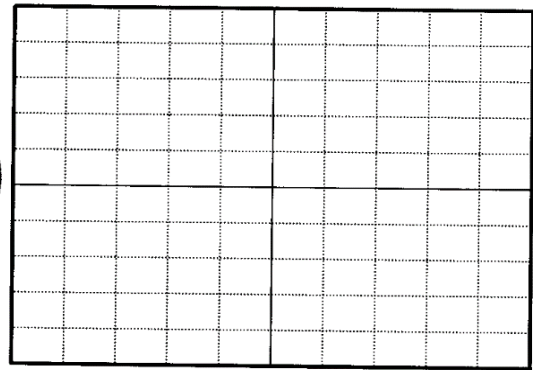
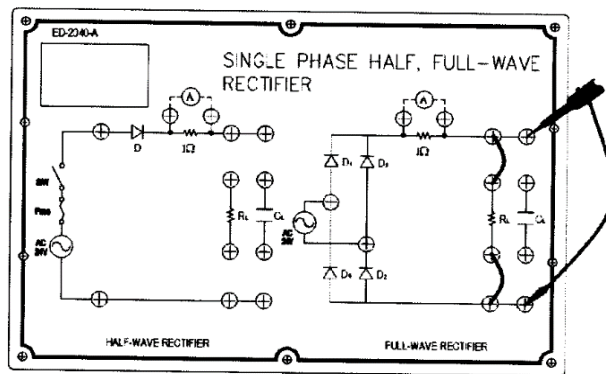
Record the results in table (II).

- The input voltage:



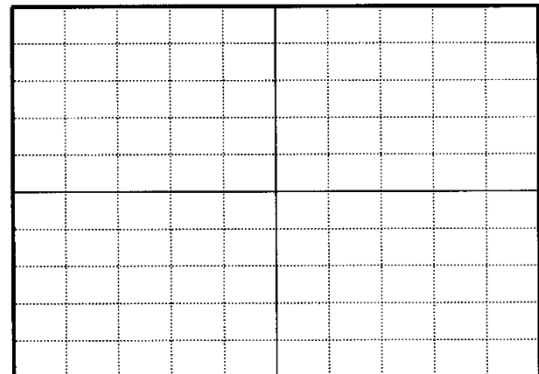
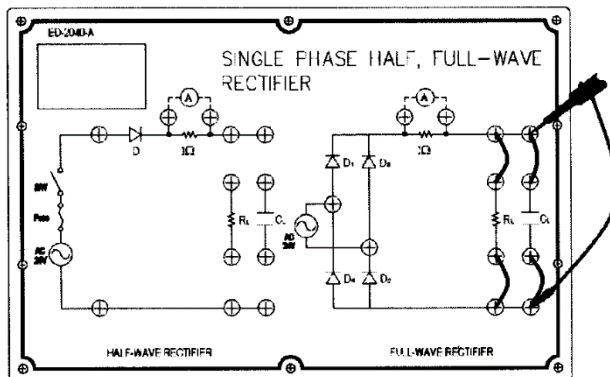
Time_____ms/div, Volt_____V/div

With R load



Time _____ ms/div, Volt _____ V/div

With RC load



Time _____ ms/div, Volt _____ V/div

Table (I) experimental results for half wave rectifier

Load Type	Value	$V_{load} (dc)$	$V_{load} (ac)$	$V_{load(rms)}$ V_{rms} $= \sqrt{(V_{dc}^2 + V_{ac}^2)}$	F.F.	R.F.	η
R	500 Ω						
R-C	500 Ω 1000 μ F						

Table (II) experimental results for full wave rectifier

Load Type	Value	$V_{load} (dc)$	$V_{load} (ac)$	$V_{load(rms)}$ V_{rms} $= \sqrt{(V_{dc}^2 + V_{ac}^2)}$	F.F.	R.F.	η
R	500 Ω						
R-C	500 Ω 1000 μ F						

Conclusion:

Laboratory Reports:

At the end of each laboratory session, each student will be required to produce their laboratory report to the lab instructor. Students are expected to prepare data, graphs and waveforms generated during their experiments. The reports are expected to include statements about their main observations of performance and characteristics of the circuit studied and their conclusions. Answers to questions for each experiment must also be included.